

Radon-Fourier Transform for Radar Target Detection, I: Generalized Doppler Filter Bank

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Based on the coupling relationship among radial velocity, range walk, and Doppler frequency of the moving target's echoes, a novel method is proposed, i.e., Radon-Fourier transform (RFT), to realize the long-time coherent integration for radar target detection. The RFT realizes the echoes spatial-temporal decoupling via joint searching along range and velocity directions, as well as the successive coherent integration via the Doppler filter bank. Besides, four equivalent RFTs are obtained with respect to the different searching parameters. Furthermore, a generalized form of RFT, i.e., generalized Radon-Fourier transform (GRFT), is also defined for target detection with arbitrary parameterized motion. Due to the similarity between the RFT and the well-known moving target detection (MTD) method, this paper provides detailed comparisons between them on five aspects, i.e., coherent integration time, filter bank structure, blind speed response, detection performance, and computational complexity. It is shown that MTD is actually a special case of RFT and RFT is a kind of generalized Doppler filter bank processing for targets with across range unit (ARU) range walk. Finally, numerical experiments are provided to demonstrate the equivalence among four kinds of RFTs. Also, it is shown that the RFT may obtain the coherent integration gain in the different noisy background and the target's blind speed effect may be effectively suppressed. In the meantime, both the weak target detection performance and the radar coverage of high-speed targets may be significantly improved via RFT without change of the radar hardware system.

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I. INTRODUCTION

It is known that pulse integration is an effective method to improve radar target detection performance in a noisy background, while the coherent integration may obtain better performance than the incoherent integration by compensating phase fluctuation among different sampling pulses [1–8]. For example, the well-known moving target detection (MTD) [1–2] method has been widely applied by modern coherent radar, where Doppler filter bank processing is used for the effective suppression of strong background clutter, as well as for the coherent integration of a moving target with unknown Doppler frequency. Besides, the MTD may be efficiently implemented via fast Fourier transform (FFT). Unfortunately, the MTD input vectors are the pulse samplings distributed along range units one-by-one and the performance gain is limited by the target's resident time in a single range unit. As a result, it is difficult to further improve the performance via MTD for the low signal-to-noise (SNR) target or the high-speed target [3–4]. For the low SNR target, e.g., far-range or low radar cross section (RCS) target, the integration time should be adequately enlarged and the range walk, i.e., the linearly varied distance between target and radar, of even a slowly moving target will exceed several range units. For a high-speed target, this kind of across range unit (ARU) effect may also be drawn even in a very short integration time. Furthermore, with the remarkably refined range resolution of modern radar, it becomes necessary to deal with the ARU effect of moving targets.

In this regard, Perry, et al. [9–10] and Zhang, et al. [11] have introduced the Keystone transform (KT) for synthetic aperture radar (SAR) ground moving targets imaging and weak radar target detection via long-time coherent integration. KT may blindly compensate the ARU effect and not destroy the pulse phase modulation. So MTD may be used for the successive coherent integration after KT. However, the existing radar, especially the searching radar, normally adopts the low pulse repetition frequency (PRF) to guarantee the far-range target detection. Therefore, most air targets may be Doppler ambiguous, for which KT may be invalidated without ambiguity correction. Yang, et al. [12] have proposed a modified KT method via simultaneous searching of the Doppler ambiguous integers and frequency, but it needs repeated implementations of high-complexity KT operators. Another more natural method is to realize the successive Doppler matching after compensation of the target's unknown range walk. Reed, et al. [13] have proposed a coherent integration method for MTD of optical image sequence. However, this method needs a 3-dimensional matched filtering, which is not realistic for radar target detection. Wang, et al. [14] have proposed an envelope compensation method

based on range stretching and time-frequency analysis, which is applicable to the time-varied Doppler case but is also computational complicated. Chen, et al. [15, 16] proposed a method to realize the echo envelopes' shifting compensation by searching the target speed and accomplish the coherent integration by using the FFT-based Doppler filter bank. However, the coupling relationship between velocity and Doppler of moving target is not exploited and its FFT-based Doppler filter bank processing seems to be abundant.

Also, there are other researchers who have proposed to adopt the incoherent integration, or the hybrid coherent-incoherent method to improve the weak target detection performance. Along this direction, the typical works may be the Hough transform (HT) based method proposed by Carlson, et al. [17–22]. The HT is used to inherently integrate the target slots exceeding the first low decision threshold, which may remarkably suppress false alarms and improve the ultimate detection performance. Furthermore, Mo, et al. [22] have also proposed an HT-based long-time integration method, which transforms the radar raw data into range-Doppler-time space and realizes the high-performance detection for targets with constant velocity or acceleration. However, without compensation of the phase fluctuation, the integration loss of the HT-based method may be large compared with a coherent integration method and it thus may not be applied when the SNR is extremely low.

Based on the signal model of a rectilinearly moving target in a long-time integration, this paper proposes a novel Radon-Fourier transform (RFT) to realize the long-time coherent integration for the moving targets with ARU range walk. Different from the existing Radon or Fourier transform methods [26, 27], RFT can effectively overcome the coupling between the range walk and phase modulations by jointly searching along range and velocity directions of moving targets. Furthermore, the Doppler filtering is also used for the successive long-time coherent integration. With different searching parameters, four equivalent continuous RFT forms are obtained. Then, the parameter space of RFTs is analyzed and discrete RFTs are also derived. Furthermore, a generalized form of RFT, called generalized RFT (GRFT), is defined for radar target detection with arbitrarily parameterized motions. Due to the similarity between the RFT and the existing MTD methods, this paper provides comparisons between them on five aspects, i.e., coherent integration time, filter bank structure, blind speed response, detection performance, and computational complexity. It is shown that the MTD is a special case of RFT method, and the RFT method is a generalized Doppler filter bank processing. Finally, some numerical experiments are provided to show that the RFT may obtain the coherent integration

gain in the different SNR background and the blind speed may also be effectively suppressed. In the meantime, both the weak target detection performance and the radar coverage of high-speed target may be significantly improved via RFT without change of the radar hardware system.

The remainder of this paper is organized as follows. In Section II the long-time integration signal model is established. In Section III we introduce the continuous RFTs. In Section IV RFT parameter design is discussed. In Section V four equivalent discrete RFTs are derived. In Section VI RFT and MTD are compared. In Section VII some numerical experiments are provided. In Section VIII conclusions are given.

II. SIGNAL MODELING FOR RADAR TARGET IN A LONG-TIME COHERENT INTEGRATION

Suppose linear frequency modulated (LFM) signal is used by radar transmitting, i.e.,

$$s_T = \text{rect}(\tau/T_p) \exp(j\pi\gamma\tau^2) \quad (1)$$

where T_p is the pulse duration, γ is the frequency rate of LFM signal, $B_s = \gamma T_p$ is the signal bandwidth. Furthermore, suppose there is a moving target with slant range R_T and radial velocity v_T at $t = 0$, then the target's range walk may be given as

$$\hat{r}(t) = R_T + v_T t, \quad t \in [-T/2, T/2]. \quad (2)$$

Normally, the target's high-order motions, e.g., acceleration, tangential motion, and so on, may exist in real targets. Fortunately, their effects on our proposed method will be reasonably controlled via limiting the coherent integration time T as discussed as Section IVA. Accordingly, the two-dimensional echoes may be represented as

$$s_r(t, \tau) = A_r \text{rect}\left(\frac{\tau - 2\hat{r}(t)/c}{T_p}\right) \exp(j\pi\gamma(\tau - 2\hat{r}(t)/c)^2) \\ \times \exp\left(-j\frac{4\pi\hat{r}(t)}{\lambda}\right), \quad t \in [-T/2, T/2] \quad (3)$$

where τ is quick-time, λ is the wavelength, c is the light speed, and A_r is the complex amplitude of the echo. After the pulse compression, the target's 2D echoes may be given as

$$s_{rm}(t, \tau) = A_{rm} \text{sinc}(\pi B_s(\tau - 2\hat{r}(t)/c)) \exp\left(-j\frac{4\pi\hat{r}(t)}{\lambda}\right), \\ t \in [-T/2, T/2] \quad (4)$$

where $A_{r,m} = T_p B A_r$. With $\hat{r} = c\tau/2$, (4) may be further rewritten as

$$s_{rm}(t, \hat{r}) = A_{rm} \text{sinc}\left(\frac{2\pi B_s(\hat{r} - \hat{r}(t))}{c}\right) \exp\left(-j\frac{4\pi\hat{r}(t)}{\lambda}\right), \\ t \in [-T/2, T/2]. \quad (5)$$

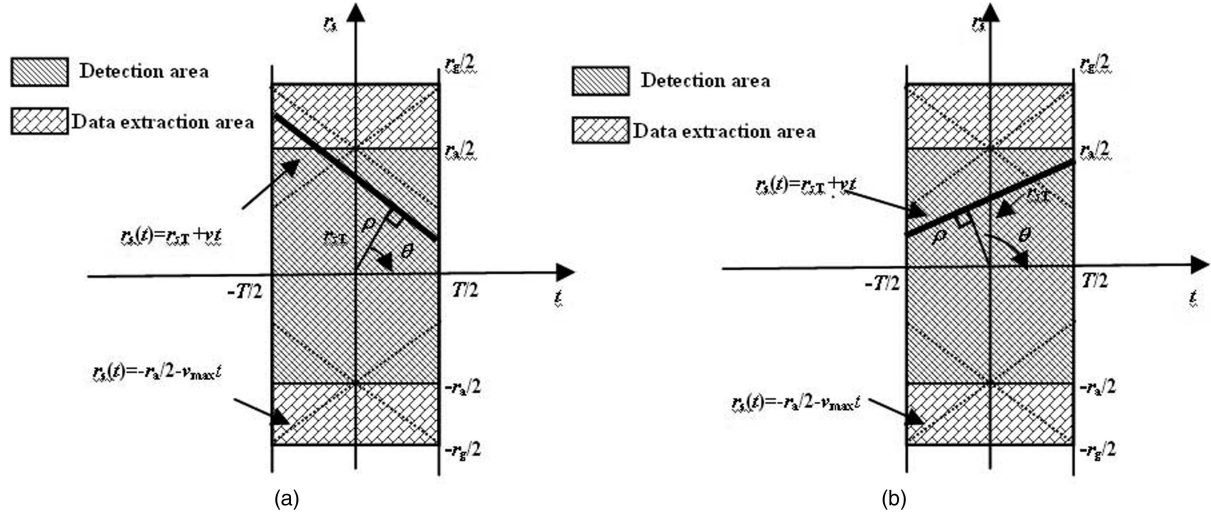


Fig. 1. Data extraction for RFT processing. (a) $\rho > 0$, $\theta \in [0, \pi/2]$ ($v < 0$, $r_{sT} > 0$). (b) $\rho > 0$, $\theta \in (\pi/2, \pi]$ ($v > 0$, $r_{sT} > 0$).

Moreover, with a pre-set scene range center $\hat{r}_c$, $\hat{r}(t)$ may be represented as

$$\hat{r}(t) = (R_T - \hat{r}_c) + \hat{r}_c + v_T t = r_{sT} + \hat{r}_c + v_T t \quad (6)$$

where the initial relative range $r_{sT} = R_T - \hat{r}_c$. Define $r_s = \hat{r} - \hat{r}_c$ the relative range and substituting (6) into (5), we have

$$s_{rm}(t, r_s) = A_T \text{sinc} \left(\frac{\pi(r_s - r_s(t))}{\rho_r} \right) \exp(j2\pi f_{dT} t), \quad t \in [-T/2, T/2] \quad (7)$$

where $r_s(t) = r_{sT} + v_T t$, $\rho_r = c/2B_s$ is the range resolution, and $A_T = A_{rm} \exp(-j4\pi(r_{sT} + \hat{r}_c)/\lambda)$ is a complex backscattering coefficient. In real applications, A_T may fluctuate and bring about phase noise for coherent integration [6, 7]. In this paper A_T is assumed to be a constant for simplicity as in many existing literatures [1–5, 8–22], and the target's Doppler is also defined as

$$f_{dT} = -2v_T/\lambda. \quad (8)$$

From (6) and (7), it is shown that the range-compressed echoes of the moving target are approximately distributed along a straight line in the $t - r_s$ plane, and Fig. 1(a)–(b) further give two cases with positive and negative radial velocities v , respectively.

III. CONTINUOUS RFT AND LONG-TIME COHERENT INTEGRATION

From Fig. 1(a)–(b), it is shown that the target's range walk curve in the $t - r_s$ plane may also be determined by another parameter pair (ρ_T, θ_T) , where polar angle $\theta_T \in [0, \pi]$ is defined as the anticlockwise angle from the range walk line to t -axis as

$$\theta_T = \text{ctg}^{-1}(-v_T). \quad (9)$$

Also, polar distance $\rho_T \in (-\infty, +\infty)$ is defined as the minimum distance between the range walk and the $t - r_s$ plane origin as

$$\rho_T = r_{sT} \sin \theta_T. \quad (10)$$

Furthermore, based on the mutual relationship between the radial velocity and $r_s(t)$ of moving target, a pair of Doppler filter functions $H_{v_T}(t)$ and $H_{\theta_T}(t)$ are define as

$$H_{v_T}(t) = \exp \left(j2\pi \frac{2v_T}{\lambda} t \right) = \exp(-j2\pi f_{dT} t) \quad (11)$$

$$H_{\theta_T}(t) = \exp \left(-j \frac{4\pi \text{ctg}(\theta_T) t}{\lambda} \right). \quad (12)$$

Then, we introduce a kind of special integration on $s_{rm}(t, r_s)$ in (7) as

$$G_{\rho\theta}(\rho_T, \theta_T) = \int_{-\infty}^{\infty} \int_{-T/2}^{T/2} s_{rm}(t, r_s) \times \delta(\rho_T - t \cos \theta_T - r_s \sin \theta_T) H_{\theta_T}(t) dt dr_s = A_T T. \quad (13)$$

It is shown that all the echoes in the integration time may be coherently integrated via (13). For comparison, we also give the Radon transform (RT) [23, 24] of $s_{rm}(t, r_s)$ as

$$R_{\rho\theta}(\rho_T, \theta_T) = \int_{-\infty}^{\infty} \int_{-T/2}^{T/2} s_{rm}(t, r_s) \times \delta(\rho_T - t \cos \theta_T - r_s \sin \theta_T) dt dr_s = \int_{-T/2}^{T/2} s_{rm}(t, r_s(t)) dt. \quad (14)$$

Obviously, we have $|G_{\rho\theta}(\rho_T, \theta_T)| \geq |R_{\rho\theta}(\rho_T, \theta_T)|$ due to the uncompensated phase changes among different pulses for (14). In real applications, because parameters r_T and v_T , as well as ρ_T and θ_T , are all unknown before target detection and parameter estimation, all the possible polar angles and polar

distances should be searched, that is

$$G_{\rho\theta}(\rho, \theta) = \int_{-T/2}^{T/2} s_{rm}\left(t, \frac{\rho}{\sin\theta} - t \operatorname{ctg}\theta\right) H_{\theta}(t) dt, \\ \rho \in [-\infty, \infty], \quad \theta \in [0, \pi] \quad (15)$$

where θ and ρ are the searching variables for polar angle and polar distance, respectively. From (9), (15) can also be rewritten as

$$G_{\rho v}(\rho, v) = \int_{-T/2}^{T/2} s_{rm}\left(t, \rho\sqrt{1+v^2} + vt\right) H_v(t) dt, \\ \rho \in [-\infty, \infty], \quad v \in [-\infty, \infty] \quad (16)$$

where v is a searching velocity. Similarly, (15) can also be rewritten as

$$G_{r\theta}(r, \theta) = \int_{-T/2}^{T/2} s_{rm}(t, r - t \operatorname{ctg}\theta) H_{\theta}(t) dt, \\ r \in [-\infty, \infty], \quad \theta \in [0, \pi] \quad (17)$$

$$G_{rv}(r, v) = \int_{-T/2}^{T/2} s_{rm}(t, r + vt) H_v(t) dt, \\ r \in [-\infty, \infty], \quad v \in [-\infty, \infty] \quad (18)$$

where r is a searching variable for slant range along the r_s -axis.

By using (15)–(18), the coherent integration outputs can be obtained with respect to the different searching parameter pairs. Only when the searching range (or polar distance) and radial velocity (or polar angle) are equal to the r_{sT} (or ρ_T) and v_T (or θ_T), the target energy distributed along multiple pulses can be coherently integrated. In other words, for the same echoes, (15)–(18) will all generate the equivalent matched “peaks,” respectively. Besides, the parameters of (ρ_T, θ_T) , (ρ_T, v_T) , (r_{sT}, θ_T) , and (r_{sT}, v_T) can be interchanged based on (9) and (10). Interestingly, (15)–(18) have the similar parts as RT or the Hough transforms (HT) [23, 24] with the additional Fourier integration component. Thus, we call (15)–(18) as RFT in this paper. Without loss of generality, the definition of RFT may be given as follows.

DEFINITION 1 Suppose a 2-D complex function $f(t, r_s) \in \mathbb{C}$ is defined in (t, r_s) plane and a parameterized line equation $r_s = \varphi(\alpha_1, \alpha_2) + \phi(\alpha_1, \alpha_2)t$ is used for searching arbitrarily lines in the plane, where $\varphi(\cdot)$ and $\phi(\cdot)$ are two specific functions to determine a line with parameters α_1 and α_2 , then the RFT is defined as $G_{\alpha_1\alpha_2}(\alpha_1, \alpha_2) = \int_{-\infty}^{\infty} f(t, \varphi(\alpha_1, \alpha_2) + \phi(\alpha_1, \alpha_2)t) \exp(j2\pi\varepsilon\phi(\alpha_1, \alpha_2)t) dt$, where ε is a known constant with respect to $f(t, r_s)$. Specifically, when the line equation $r_s = r + vt$ is used, the transform $G_{rv}(r, v) = \int_{-\infty}^{\infty} f(t, r + vt) \exp(j2\pi\varepsilon vt) dt$ is called “standard RFT” due to the explicit meaning of parameter r and v .

By comparing the definitions of RFT and the existing RT or HT, e.g., (14) [23, 24], there are some

important differences among them, three of which are given as follows.

1) The existing RT and HT, especially HT, are normally defined in the real image domain regardless of coherent integration of the phase information, which are normally used to enforce the special geometry structures, e.g., line, curve, or surface, in the image by exploiting the correlation on the signal’s amplitude. The proposed (15)–(18) are defined in the complex signal domain, which jointly exploits the amplitude and phase information.

2) For the integration transforms as (15)–(18), the important Doppler compensation functions $H_v(t)$ or $H_{\theta}(t)$ are introduced to act as a Fourier integration component, which cancels the phase fluctuations among different pulses and generates the ultimate coherent “peak.” The introduction of these operators is the main differences between the proposed method and the existing HT or RT methods. In this sense, the RFT is applying Fourier integration along a line with a certain direction, which is determined via the RT. In Section VI, it is proven that RFT has similar effects as the conventional Doppler filter bank, e.g., the MTD method, but it will significantly outperform the MTD via a relatively long integration time.

3) In (16)–(18) the slant range r or radial velocity v is involved, which all have clear physical meaning in radar signal processing, while the transform parameters of conventional RT or HT normally only have the geometry structure meanings but no explicit physical meaning or background.

Furthermore, based on the proposed RFT as (15)–(18), we further give a more generalized long-time coherent integration method, i.e., GRFT, for radar MTD with any parameterized motion as follows.

DEFINITION 2 Suppose a 2-D complex function $f(t, r_s) \in \mathbb{C}$ is defined in (t, r_s) plane and a parameterized $N + 1$ dimensional equation $r_s = \eta(\alpha_1, \dots, \alpha_N, t)$ is used for a certain time-varied curve in the plane, then the GRFT is $G(\alpha_1, \dots, \alpha_N) = \int_{-\infty}^{\infty} f(t, \eta(\alpha_1, \dots, \alpha_N, t)) \exp(2\pi\varepsilon\eta(\alpha_1, \dots, \alpha_N, t)) dt$ where ε is a known constant with respect to $f(t, r_s)$.

Why do we call the proposed method GRFT? The reason is that the part $\int_{-\infty}^{\infty} f(t, \eta(\alpha_1, \dots, \alpha_N, t)) dt$ in the GRFT may be regarded as a generalized RT along a parameterized curve and the part $\exp(2\pi\varepsilon\eta(\alpha_1, \dots, \alpha_N, t))$ in the GRFT actually acts as a phase compensation component for coherent integration. Obviously, the RFT in Definition 1 is a special case of GRFT when the searching parameterized curve is degenerated into a line and the certain phase $\exp(j2\pi\varepsilon\varphi(\alpha_1, \alpha_2))$ is omitted for simplicity. More importantly, by using the proposed GRFT processing, coherent radar’s moving target echoes with any parameterized range migration may be coherently integrated in any long coherent time.

IV. PARAMETER DESIGN FOR RFT IN REAL RADAR APPLICATION

To apply (15)–(18) in real radar target detection, let's define the minimum RCS of the target as $\sigma_{\min}$. The maximum positive target's radial velocity is $v_{\max}$. That is, the radial velocity of the target is distributed in the area of $[-v_{\max}, v_{\max}]$. The maximum target's radial acceleration is $a_{\max}$. Furthermore, define the detecting range area as $[r_c - r_a/2, r_c + r_a/2]$. That is, all the targets to be detected are distributed in the range area r_a around center r_c .

A. Determine the Coherent Integration Time

1) *Considerations of the Coherent Integration Gain:* Based on the given radar system parameters, the radar equation may be used to obtain the minimum SNR (in dB) of the raw data for the weakest target in the farthest range as

$$\text{SNR}_{\min} = 10\log_{10}\left(\frac{P_t G^2 \lambda^2 \sigma_{\min}}{(4\pi)^3 k T_s B F L R_{\max}^4}\right) \quad (19)$$

where k is the Boltzmann constant, T_s is the system Kelvin temperature, F is the noise coefficient, B is the system bandwidth, L is the system loss, and $R_{\max} = r_c + (r_a + v_{\max}T)/2 \approx r_c + r_a/2$ is the maximum range of the range gate. Then, the minimum SNR after pulse compression may be given as

$$\text{SNR}_{\min, \text{PC}} = 10\log_{10}\left(\frac{P_t G^2 \lambda^2 \sigma_{\min} D}{(4\pi)^3 k T_s B F L R_{\max}^4}\right) \quad (20)$$

where $D = B_s T_p$ is the time bandwidth product (TBP) of the transmitting signal, which shows that the ideal SNR gain of pulse compression may be

$$G_{\text{PC}} = \text{SNR}_{\min, \text{PC}} = 10\log_{10}(D). \quad (21)$$

To effectively detect the target in noisy background, high enough SNR should be needed [4]. For example, to obtain the detection probability $P_d = 80\%$ with a constant false alarm probability $P_f = 10^{-6}$ in the additive white Gaussian noise (AWGN) background, the SNR should be roughly larger than 12.8 dB. Define the required minimum SNR of correct detection as SNR_{req} . Then, the needed SNR gain should be obtained via coherent integration as

$$\begin{aligned} \text{SNR}_{\text{CIT}} &= \text{SNR}_{\text{req}} - \text{SNR}_{\min, \text{PC}} \\ &= \text{SNR}_{\text{req}} - \text{SNR}_{\min} - G_{\text{PC}} \end{aligned} \quad (22)$$

because

$$\text{SNR}_{\text{CIT}} = 10\log_{10}(N_a) = 10\log_{10}(T f_p) \quad (23)$$

where f_p is system PRF. So the coherent integration time may also be determined from (23) as

$$T_{\text{CIT}} = 10^{(\text{SNR}_{\text{CIT}}/10)}/f_p. \quad (24)$$

2) *Consideration of Target Acceleration:* It is easy to see that the derived RFT is based on the assumption of the target's uniformly radial velocity. However, many real targets have radial acceleration or high-order motion. Besides, the uniform tangential motion will also bring about the change of relative radial velocity. Normally, either the direct radial acceleration or the uniform tangential motion may generate the fixed or unfixed radial acceleration for RFT, which will cause both Doppler and envelope mismatches between the real echoes and the searching parameters of RFT. So the coherent integration gain will be inevitably decreased, accordingly.

Assume the time-varied acceleration of the target in the whole integration time is $a(t)$, and the target's range migration $r_T(t)$ can be approximated as

$$r_T(t) = (r_T - r_c) + v_T t + \frac{1}{2}a(t)t^2 = r_{sT} + v_T t + \frac{1}{2}a(t)t^2, \quad t \in [-T/2, T/2]. \quad (25)$$

Then, the instantaneous Doppler frequency may be given as

$$f_d(t) = -\frac{2}{\lambda} \frac{dr_T(t)}{dt} = -\frac{2v_T + 2a(t)t}{\lambda}, \quad t \in [-T/2, T/2]. \quad (26)$$

In order to guarantee coherent integration performance, we limit the Doppler change in coherent integration time to be less than a Doppler resolution ρ_d , i.e.,

$$\Delta f_d(t)|_{t \in [-T/2, T/2]} \leq |2a_{\max}T/\lambda| < \rho_d = 1/T. \quad (27)$$

So we have the constraint on coherent time as

$$T_{a, \text{Doppler}} < \sqrt{\lambda/(2a_{\max})}. \quad (28)$$

Besides, the acceleration will also bring about the range curvature as (25). Because the RFT cannot compensate the nonlinear range curvature, the range curvature should be limited as

$$\frac{1}{2}a_{\max}t^2 \Big|_{t=\pm T/2} = \frac{a_{\max}T^2}{8} < \rho_r \quad (29)$$

and

$$T_{a, \text{curvature}} < \sqrt{8\rho_r/a_{\max}}. \quad (30)$$

Normally we have $\rho_r \gg \lambda$ for radar, and (28) is a more strict condition than (30).

3) *Consideration of the Illumination Time:* Obviously, the real radar illumination time of a target is also an important factor to affect the coherent integration performance. In the conventional mechanical scanning radar with single beam, the target illumination time is seriously limited due to the fast rotation of the radar antenna. The coherent integration time will be limited for RFT, accordingly, but not for the new-generation radar [25] with phased array antenna, which normally abandons the mechanical

scanning and adopts the multiple fixed beams to cover the detecting area. Thus, the target illumination time may be enlarged to a great extent.

Assuming the real illumination time is T_i and combining the above three factors, we have

$$T = \min(T_{\text{CIT}}, T_{a, \text{Doppler}}, T_i) \quad (31)$$

where $\min(\cdot)$ is the minimize operator. In the real application, when the prior knowledge of a_{max} is unavailable, we also can determine coherent integration time as

$$T = \min(T_{\text{CIT}}, T_i). \quad (32)$$

Then, compared with uniform moving target, the loss of SNR gain should be considered for targets with acceleration larger than

$$a_T = \lambda^2 / (2T)^2. \quad (33)$$

B. Determine the Range Sampling Gate

As in Fig. 1, in order to guarantee all moving targets can be completely integrated during the coherent integration time, the range sampling gate r_g should be selected as

$$r_g = r_a + v_{\text{max}} T. \quad (34)$$

Obviously, we have $r_g \geq r_a$ according to the v_{max} definition and r_g should be increased with the increase of coherent integration time T .

C. Determine the Parameter Space of RFT

In accordance with the searching velocity area $[-v_{\text{max}}, v_{\text{max}}]$ and searching range area $[-r_g/2, r_g/2]$, (15)–(18) can be rewritten with a reasonable limited parameter space as

$$G_{\rho\theta}(\rho, \theta) = \int_{-T/2}^{T/2} s_{rm}(t, \rho / \sin \theta - t \text{ctg} \theta) H_{\theta}(t) dt$$

$$\rho \in \left[-\frac{r_a}{2\sqrt{1+v_{\text{max}}^2}}, \frac{r_a}{2\sqrt{1+v_{\text{max}}^2}} \right],$$

$$\theta \in [\text{ctg}^{-1}(v_{\text{max}}), \text{ctg}^{-1}(-v_{\text{max}})] \quad (35)$$

$$G_{\rho v}(\rho, v) = \int_{-T/2}^{T/2} s_{rm}(t, \rho \sqrt{1+v^2} + vt) H_v(t) dt$$

$$\rho \in \left[-\frac{r_a}{2\sqrt{1+v_{\text{max}}^2}}, \frac{r_a}{2\sqrt{1+v_{\text{max}}^2}} \right],$$

$$v \in [-v_{\text{max}}, v_{\text{max}}] \quad (36)$$

$$G_{rv}(r, v) = \int_{-T/2}^{T/2} s_{rm}(t, r - t \text{ctg} \theta) H_{\theta}(t) dt$$

$$r \in [-r_a/2, r_a/2],$$

$$\theta \in [\text{ctg}^{-1}(v_{\text{max}}), \text{ctg}^{-1}(-v_{\text{max}})] \quad (37)$$

$$G_{rv}(r, v) = \int_{-T/2}^{T/2} s_{rm}(t, r + vt) H_v(t) dt$$

$$r \in [-r_a/2, r_a/2], \quad v \in [-v_{\text{max}}, v_{\text{max}}]. \quad (38)$$

Obviously, for the different searching velocity or polar angle, the searching polar distance should also be varied.

D. Determine Searching Interval and Number for Radial Velocity and Polar Angle

It is obvious that the key step of RFT different from RT of HT is the Doppler integration components as (11) and (12). After selected coherent integration time T , the Doppler frequency resolution of Doppler filter may be given as

$$\rho_d = 1/T. \quad (39)$$

Therefore, according to (8), the searching velocity interval may be determined as

$$\Delta_v = \lambda/2T. \quad (40)$$

Furthermore, the searching velocity number is

$$N_v = \text{round}(2v_{\text{max}}/\Delta_v) \quad (41)$$

where $\text{round}(\cdot)$ denotes the integer operator. Normally, when the (35) and (37) are used for RFT processing, the number of searching polar angle can be determined the same as the searching radial velocity. Therefore, the searching polar angle interval may be rewritten from (9) as

$$\Delta_{\theta, \tilde{v}} = \text{ctg}^{-1}(\tilde{v} + \Delta_v) - \text{ctg}^{-1}(\tilde{v}) \quad (42)$$

where $\tilde{v}$ is the current searching radial velocity, and it is shown that the polar angle searching interval should be not a constant but a function of $\tilde{v}$.

E. Determine Searching Interval and Number for Range and Polar Distance

When (38) is used for RFT in the parameter space (r, v) , the range sampling frequency f_s may also be determined according to the signal bandwidth as

$$f_s \geq \gamma T_p. \quad (43)$$

The searching interval in range can also be as large as the range-compressed data, i.e.,

$$\Delta_r = 2f_s/c \leq 2\gamma T_p/c. \quad (44)$$

Therefore, the number of searching range is

$$N_r = \text{round}(r_a/\Delta_r). \quad (45)$$

Conventionally, the searching polar distance number can also be set as N_r in (37) and (38). From (9), the searching polar distance interval of RFT is determined

as

$$\Delta_\rho = \Delta_r \sin(\theta) = \frac{2\gamma T_p}{\sqrt{1+v^2}}. \quad (46)$$

It is obvious that the searching interval changes with searching radial velocity v or polar angle θ . But after the determination of θ or v , the Δ_ρ is also determined.

V. DISCRETE RFT TRANSFORM

From the radar system PRF f_p and range sampling frequency f_s , the discrete signal in the integration time T and range gate r_g may be represented as

$$s_{rm}(m, n) \quad m = -N_a/2, \dots, N_a/2 - 1, \quad n = 1, \dots, N_g \quad (47)$$

where $N_a = T f_p$ denotes the integration pulse number, and the range unit number in the range gate is given as

$$N_g = \text{round}(f_s(2r_g/c + T_p)). \quad (48)$$

Based on the above pulse discrete sampling, (11) and (12) may be rewritten as

$$H_v(m) = \exp\left(j \frac{4\pi v}{\lambda} m T_r\right), \quad m = 1, \dots, N_a \quad (49)$$

$$H_\theta(m) = \exp\left(-j \frac{4\pi \text{ctg}(\theta)}{\lambda} m T_r\right), \quad m = 1, \dots, N_a \quad (50)$$

where $T_r = f_p^{-1}$ is the pulse repetition interval. Based on the proposed sampling areas and intervals of radial velocity, polar angle, range, and polar distance for (39)–(46), we further define the discrete series of searching radial velocity, polar angle, range and polar distance as

$$v(q) = -v_{\max} + q\Delta_v, \quad q = 1, \dots, N_v \quad (51)$$

$$\theta(k) = \text{ctg}^{-1}(-v(k)), \quad k = 1, \dots, N_v \quad (52)$$

$$r_s(i) = -r_a/2 + i\Delta_r, \quad i = 1, \dots, N_r \quad (53)$$

$$\rho_k(l) = r(l) \sin(\theta(k)), \quad l = 1, \dots, N_r, \quad k = 1, \dots, N_v. \quad (54)$$

From (51)–(54), (35)–(38) may be rewritten as

$$G_{\rho\theta}(l, k) = \sum_{m=-N_a/2}^{N_a/2-1} s_{rm} \left(m, \text{round} \left(\frac{\rho_k(l) - m T_r \cos(\theta(k))}{\Delta_r \sin(\theta(k))} \right) \right) \times H_{\theta(k)}(m) \quad l = 1, \dots, N_r, \quad k = 1, \dots, N_v \quad (55)$$

$$G_{rv}(i, q) = \sum_{m=-N_a/2}^{N_a/2-1} s_{rm} \left(m, \text{round} \left(\frac{\rho_q(l) \sqrt{1+v(q)^2} + m T_r v(q)}{\Delta_r} \right) \right) \times H_{v(q)}(m) \quad l = 1, \dots, N_r, \quad q = 1, \dots, N_v \quad (56)$$

$$G_{r\theta}(i, k) = \sum_{m=-N_a/2}^{N_a/2} s_{rm} \left(m, \text{round} \left(\frac{r(i) - m T_r \text{ctg}(\theta(k))}{\Delta_r} \right) \right) \times H_{\theta(k)}(m) \quad i = 1, \dots, N_r, \quad k = 1, \dots, N_v \quad (57)$$

$$G_{rv}(i, q) = \sum_{m=-N_a/2}^{N_a/2-1} s_{rm} \left(m, \text{round} \left(\frac{r(i) + m T_r v(q)}{\Delta_r} \right) \right) H_{v(q)}(m) \quad i = 1, \dots, N_r, \quad q = 1, \dots, N_v. \quad (58)$$

Obviously, (55)–(58) are the discrete RFT forms of (15)–(18). Notably, the operator $\text{round}()$ in (55)–(58) is only used for simplicity, the interpolation or the FFT-based fractal processing may be used in real application for obtaining higher transform accuracy.

VI. COMPARISONS BETWEEN RFT AND MTD

All the proposed (55)–(58) may be decomposed into two subsequent steps. First, extract the samplings with respect to certain searching radial velocity (or polar angle) and searching range (or polar distance). Second, accomplish the coherent integration via the Fourier integration with Doppler function as (11) or (12). Because there are a total of N_v searching radial velocity (or polar angles) to be used, the RFT method is actually to use the Doppler filter bank processing. As we know, the well-known MTD method in the radar signal processing is also a Doppler filter bank processing to realize clutter suppressing, as well as the moving target coherent integration. What are the differences between MTD and the proposed RFT? To answer the question, we provide detailed comparisons between them on five aspects, i.e., coherent integration time, filter bank structure, blind speed response, integration performance, and computational burden, respectively.

A. Comparison of Coherent Integration Time

As illustrated in Fig. 2, the conventional MTD method limits the slant-range changes of all the detecting targets less than a range unit in the coherent integration time T_M , i.e.,

$$T_M \leq \rho_r / v_{\max}. \quad (59)$$

Then, the Doppler filter bank is adopted to match the target's echoes in T_M with different radial velocity. Different from (59), the integration time of RFT is unrelated to the target radial velocity and range resolution, but related to the target's high-order motion, illuminating time and the needed SNR gain as (31). Therefore, many more pulses can be coherently integrated for RFT in a longer time as Fig. 2, which may give a better solution for the following five applications of modern radar.

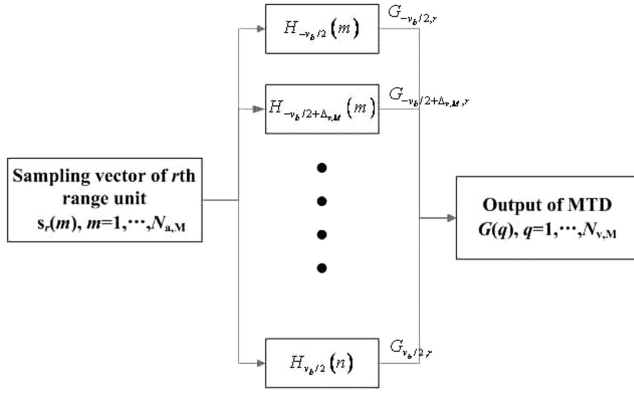


Fig. 3. Filter bank structure of MTD processing.

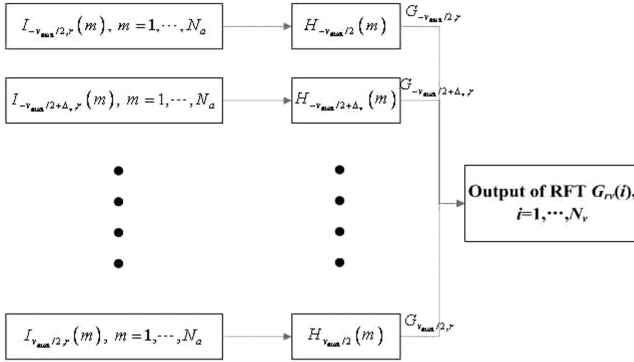


Fig. 4. Filter bank structure of RFT processing.

functions of RFT are also periodically changed with the searching velocity (or polar angle), the input series are different for different filters due to the extracted data as (55)–(58). On the other hand, because the velocity covering area is $[-v_{\max}, v_{\max}]$ and the searching velocity interval and filter number are illustrated as (40) and (41), the filter order and filter number of RFT processing are normally different for RFT. The filter bank structures of MTD and discrete RFT are given as Fig. 3 and Fig. 4, respectively.

In the Fig. 4, the input series $I_{v,r}(m)$ is

$$I_{v,r}(m) = s_{rm}(m, \text{round}(r + mT_r v)), \quad m = 1, \dots, N_a. \quad (65)$$

From Fig. 4 it is shown that the input series of RFT filters are different with the different searching velocities. The difference among different series may become more and more evident with the improvement of the range resolution and the increase of coherent integration time. Conversely, if the coherent integration time is confined to T_M as in Fig. 2, then all the inputs of the RFT filter may be identical and the RFT is degenerated into the MTD processing. Therefore, the MTD method may be regarded as a special case of the RFT processing. Second, the filter number of RFT is normally larger than the filter number of MTD, i.e., $N_v > N_{v,M}$. Third, as for the filter design, the filter order of RFT is also larger than MTD, i.e., $N_a > N_{a,M}$.

C. Comparison of Blind Speed Response

For the existing MTD processing, the coherent integration time T_M should be constrained so that the range migrations of all moving targets will not exceed a certain range unit. On the other hand, the Doppler filter responses are periodically repeated along the searching radial velocity with a certain cycle v_b . So for the filter bank of MTD with searching area of $[-v_b/2, v_b/2]$, the responses for the targets with velocity $v_T + pv_b$ will be identical, where p is an uncertain integer. That is, after the MTD filtering, the target's true radial velocity v_T cannot be ascertained and velocity is still ambiguous. To solve this problem, the PRF staggering technique [1–2] has been widely adopted by modern radar at the cost of system complexity increase and integration performance reduction. But for RFT, the ambiguity of blind speed may be well suppressed, because the input vectors of different filters are different with respect to different searching parameters. Thanks to the cut-off property of Doppler filter, the output will approach zero when velocity mismatch is larger than the velocity resolution. Even when the velocity mismatch is exactly integer times of blind speed, the outputs of RFT will also be effectively suppressed because the range walks of two kinds of velocity are different.

Here, we provide a simple interpretation for the blind speed response of discrete RFT via (58). At first, a prior condition is defined as

$$T > \rho_r / v_b \quad (66)$$

which is also a requirement to suppress blind speed response via RFT processing. Otherwise, at least the response of $v_T \pm v_b$ will have an identical response as v_T and a pair of one times of “blind speed sidelobe (BSSL)” will be generated for RFT. Fortunately, (66) is easily satisfied for conventional radar. With the prior condition as (66), when the searching radial velocity is $v_T + pv_b$, then the searching range walk may have an “intersection point” with the real target range walk, and the pulse number covered by the “intersection point” is

$$N_{a,p} \approx \frac{f_p \rho_r}{pv_b}. \quad (67)$$

Because these $N_{a,p}$ pulse samplings are also matched to the $v_T + pv_b$ filter due to the periodical repetition of the Doppler filter function, the output for the searching velocity $v_T + pv_b$ will also coherently integrate these $N_{a,p}$ pulse samplings and generate a p th BSSL. Furthermore, primary lobe-to-sidelobe ratio (PSLR) of p th BSSL can be obtained as

$$I_p = \frac{N_a}{N_{a,p}} = \frac{pN_a \lambda}{2\rho_r}. \quad (68)$$

Also, the generalized proof of (68) may be found in [28]. From (65), it is shown that the PSLR of the p th BSSL may be improved with the increase of p , N_a , λ , and B_s . In other words, with a prior condition (66), the RFT may be used to suppress the BSSL. The larger p is, the better PSLR may be obtained.

D. Comparison of Detection Performance

For a nonfluctuating target with constant RCS σ , maximum radial velocity $v_{\max}$ and range R , the output SNR may be calculated based on the different possible integration time for MTD processing, Hough-based method [23, 24], and RFT, respectively,

$$\text{SNR}_{\text{MTD}} = \frac{P_t G^2 \lambda^2 D N_{a,\text{MTD}} \sigma}{(4\pi)^3 k T B F L R^4} \quad (69)$$

$$\text{SNR}_{\text{Hough}} = \frac{P_t G^2 \lambda^2 D \sqrt{N_a N_{a,\text{MTD}}} \sigma}{(4\pi)^3 k T B F L R^4} \quad (70)$$

$$\text{SNR}_{\text{RFT}} = \frac{P_t G^2 \lambda^2 D N_a \sigma}{(4\pi)^3 k T B F L R^4}. \quad (71)$$

Formulas (69) and (70) are obtained with the coherent integrated pulse number and by neglecting the data extraction error of discrete RFT. Meanwhile, the calculation process of the Hough-based method is given as follows. Suppose that the N_a coherent integrated pulse may be divided into N_s nonoverlapped subapertures, each of which contains $N_{a,M}$ coherent integrated pulses. Then we have

$$N_s = N_a / N_{a,M}. \quad (72)$$

The Hough-based method may obtain the SNR gain $N_{a,M}$ in each subaperture, and obtain the incoherent integration gain among the N_s subapertures, and the SNR gain may be approximated as $\sqrt{N_s}$. Therefore, the total integration gain via the hybrid coherent and incoherent integration may be calculated as $\sqrt{N_a N_{a,M}}$.

Similarly, according to (69)–(71), the minimum detectable RCS of a nonfluctuating target at range R_T may be given as

$$\sigma_{\min,\text{MTD}} = \frac{(4\pi)^3 k T_s B F L R_T^4 \text{SNR}_{\min}}{P_t G^2 \lambda^2 D N_{a,M}}, \quad T > T_M \quad (73)$$

$$\sigma_{\min,\text{Hough}} = \frac{(4\pi)^3 k T_s B F L R_T^4 \text{SNR}_{\min}}{P_t G^2 \lambda^2 D \sqrt{N_a N_{a,M}}}, \quad T > T_M \quad (74)$$

$$\sigma_{\min,\text{RFT}} = \frac{(4\pi)^3 k T_s B F L R_T^4 \text{SNR}_{\min}}{P_t G^2 \lambda^2 D N_a}, \quad T > T_M. \quad (75)$$

According to (69)–(71), the maximum detecting range may be given for three methods as

$$R_{\max,\text{MTD}} = \left(\frac{P_t G^2 \lambda^2 \sigma D N_{a,M}}{(4\pi)^3 k T_s B F L \text{SNR}_{\min}} \right)^{1/4} \quad (76)$$

TABLE I
Complexity of MTD and RFT

Methods	FIR-Based	FFT-Based
RFT	$N_r N_v (N_a I_m + (N_a - 1) I_a)$	Under the development
MTD	$N_r N_{v,M} \left(\frac{N_{v,M} I_m}{2} + (N_{v,M} - 1) I_a \right)$	$N_r \left(\frac{N_{v,M} \log_2(N_{v,M}) I_m}{2} + N_{v,M} \log_2(N_{v,M}) I_a \right)$

Note: I_m : complex multiplication, I_a : complex addition, FIR: finite impulse response.

$$R_{\max,\text{Hough}} = \left(\frac{P_t G^2 \lambda^2 \sigma D \sqrt{N_a N_{a,M}}}{(4\pi)^3 k T_s B F L \text{SNR}_{\min}} \right)^{1/4} \quad (77)$$

$$R_{\max,\text{RFT}} = \left(\frac{P_t G^2 \lambda^2 \sigma D N_a}{(4\pi)^3 k T_s B F L \text{SNR}_{\min}} \right)^{1/4}. \quad (78)$$

Comparing MTD, RFT, and Hough-based methods, the proposed RFT method may detect the weakest target at the same range, or detect the same target at the farthest range.

E. Comparisons of Computational Complexity

It should be pointed out that the computational complexity of RFT may be much larger than MTD. Specifically, the complexity increase is caused by the following four factors.

1) Because the integration time of RFT is larger than the MTD processing, i.e., $T > T_M$, the filter order of RFT is normally larger than MTD, i.e., $N_a > N_{a,M}$.

2) Because the searching velocity area of RFT $[-v_{\max}, v_{\max}]$ is normally larger than that of MTD $[-v_b/2, v_b/2]$, the RFT filter number is also larger than MTD, i.e., $N_v > N_{v,M}$.

3) For the MTD processing, because the input series is an identical series, the successive filter bank may be jointly realized via the FFT operator to further reduce the computational burden, while the RFT method can not directly use the FFT-based filter bank because the input series are different for different filters.

4) For the RFT processing as (55)–(58), the addressing operator and data extraction should be implemented to obtain the inputs for different filters. However, because the addressing principle for RFT is certain when the radar system parameters are given, it is possible to reduce and control the addressing burden by combining the hardware design, and the addressing operator is omitted for discussion in this paper.

To quantitatively compare the complexities between the MTD and RFT, the computational amounts of two kinds of filter bank processing are given as Table I.

By combining the differences between RFT processing and MTD on the above five aspects, i.e., coherent integration time, filter band structure, blind

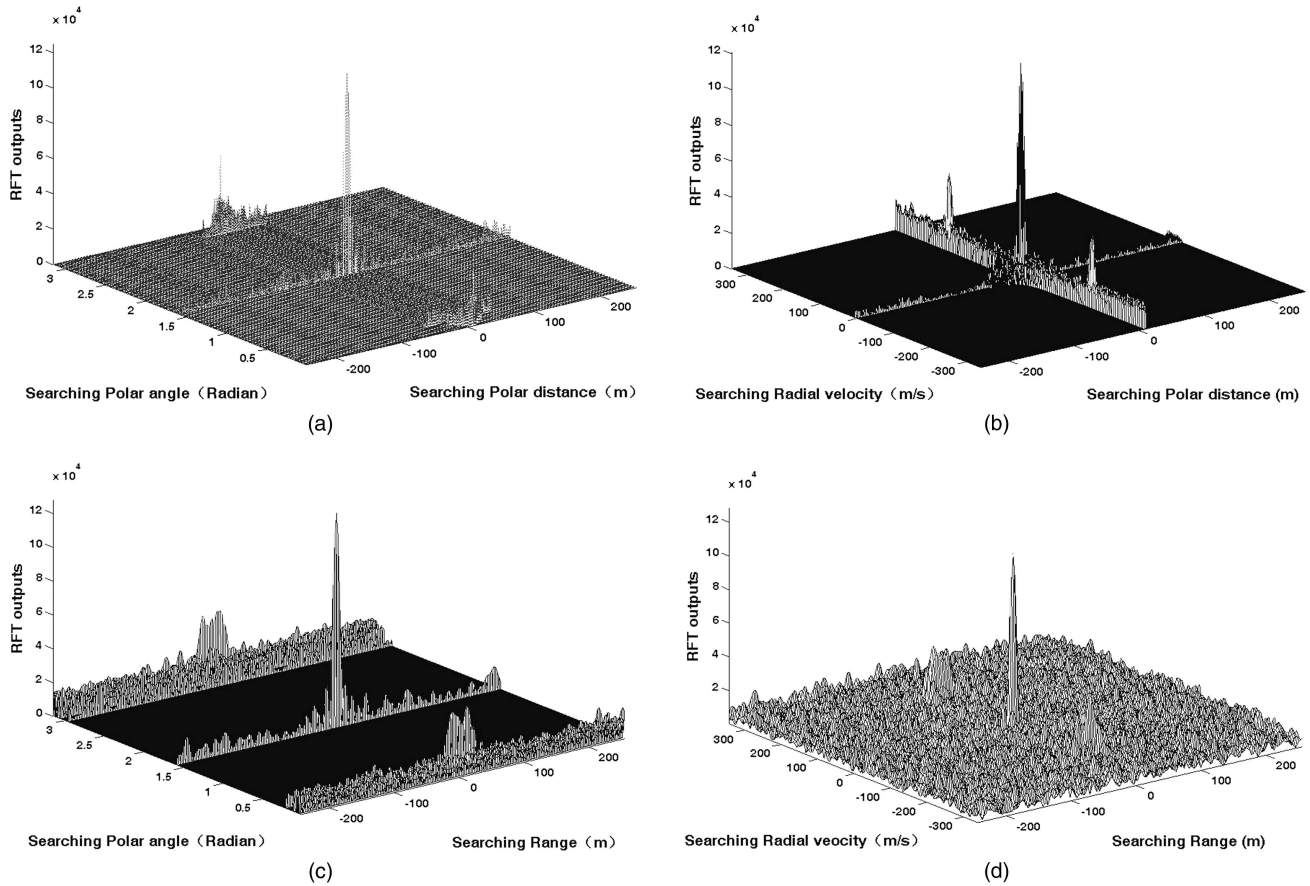


Fig. 5. Equivalence of four discrete RFT methods. (a) RFT results as (55) with $r_c = 150$ Km. (b) RFT results as (56) with $r_c = 150$ Km. (c) RFT results as (57) with $r_c = 150$ Km. (d) RFT results as (58) with $r_c = 150$ Km.

speed response, and computational complexity, some significant conclusions can be given as follows.

- 1) MTD is a special case of RFT when the coherent integration time is limited. Also, the RFT processing is a generalized MTD for target with a linear ARU effect, and it may significantly improve detection performance for low-SNR and high-speed targets.
- 2) Because RFT is also a kind of Doppler filter bank processing, it is straightforward that it will effectively improve MTD in the static or slow moving clutter as MTD. The performance analysis is omitted for simplicity in this paper.
- 3) The RFT method may effectively suppress the blind speed as discussed in Section VIC and its proofs may be found in [28].
- 4) Because longer integration time is used, the higher estimated velocity resolution and accuracy of target may be obtained via the RFT method than the MTD-based method [28].
- 5) Because the computational burden of RFT is normally larger than MTD, it is necessary to propose a fast algorithm for RFT.

VII. NUMERICAL EXPERIMENTS

To verify the proposed RFT method, this section gives some detailed numerical experiments. At

first, the parameters of an experimental searching radar are given as follows: Transmitting peak power $P_t = 100$ Kw, transmitting LFM pulse duration $T_p = 50$ us, bandwidth $B_s = 15$ MHz, antenna gain $G = 24$ dB, radar carrier $f_c = 150$ MHz, the IF bandwidth $B_{IF} = 17$ MHz, $T_s = 290$ K, noise coefficient $F_n = 3$ dB, azimuth beam width $|\theta_\beta| = 9^\circ$, $f_p = 200$ Hz, $f_s = 60$ MHz, and system loss $L = 15$ dB (with consideration of feeder transferring loss, receiver mismatch loss, the nonrectangular antenna beam loss, target fluctuation, and so on).

A. Equivalence of Four Discrete RFTs

Suppose there is target with $\sigma_T = 1$ m², $R_T = 150$ Km, and 0 m/s radial velocity. The maximum searching velocity is $v_{\max} = 1$ Mach = 340 m/s. In the coherent integration time $T = 0.2$ s, the experimental results of four discrete RFT transform as (55)–(58) with $r_c = 150$ Km are given as Fig. 5(a)–(d), respectively. It is shown that the peaks of four RFT processing are all equal to $A = D \times N_a \times f_s / B = 12 \times 10^4$, which validates four discrete RFT methods with equivalent coherent integration performances.

It is noted that the searching polar angles are extremely discontinuously distributed as Fig. 5(a) or Fig. 5(c), which is caused by the nonlinear relationship between the radial velocity and polar

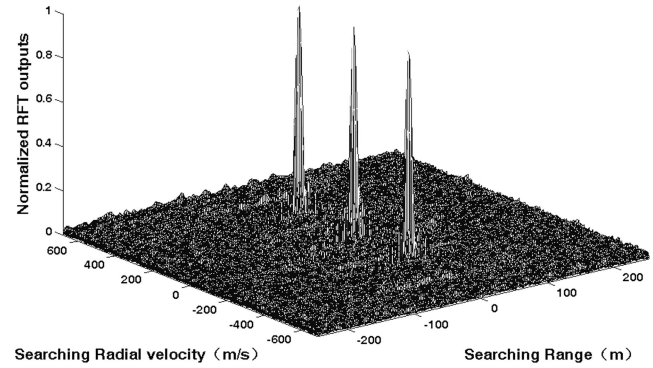
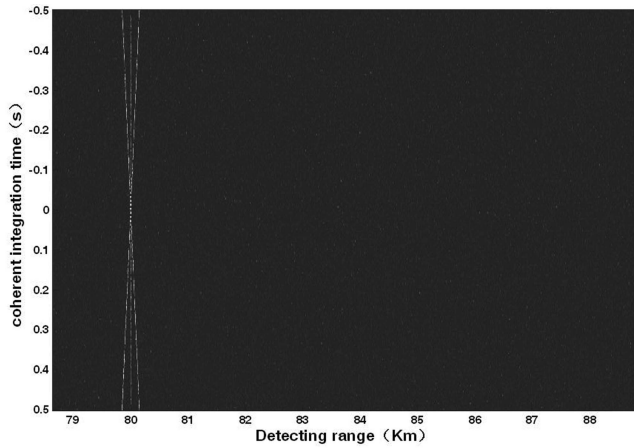


Fig. 6. Detection at $R_T = 80$ Km. (a) After pulse compression. (b) RFT with $r_c = 80$ Km.

angle. Among the four RFTs as (55)–(58), the standard discrete RFT as (58) has the explicit physical meanings and its searching peak corresponds to the target radial velocity and range, respectively. Notably, the physical range r_{real} is a shift of “searching range” as $r_{\text{real}} = r_c + r$. Furthermore, because the parameter space of standard discrete RFT as (58) is two-dimensional uniformly distributed, which is suitable for the successive constant false alarm ratio (CFAR) detection. Interestingly, it can be found that there are some subpeaks distributed along the searching velocity or polar angle as Fig. 5(a)–(d) and they are exactly located at the blind speed ± 200 m/s. These subpeaks are just the so-called BSSL, whose properties are discussed in [28].

B. Detection Performance of RFT Method in Different SNR Background

To verify the SNR gain and the detection performance of the RFT method for the targets with different slant range, experimental radar with phased array antenna is designed in this subsection. This radar adopts azimuth multiple beams to cover whole scanning space. The other system parameters are identical to Section VIIA. Furthermore, three targets with the same backscattering 1 m^2 RCS are added to the simulation scene. At $t = 0$, these three targets are located at the same range unit, and their radial velocities are 0 m/s, 300 m/s and -300 m/s, respectively. The coherent integration time is $T = 1$ s and the outputs of pulse-compression and RFT transforms are provided when targets are located at 80 Km, 150 Km, and 300 Km, respectively.

1) $R_T = 80$ Km: When the target is located at 80 Km, the SNR of the targets’ raw data are -11.0 dB based on the known system parameters. Because the time-bandwidth product $D = 750$, so the SNR gain of pulse compression is $G_{\text{pc}} = 10\log_{10}(D) = 28.8$ dB and the targets are detectable after the pulse compression as Fig. 6(a). Furthermore,

with 23 dB coherent integration SNR gain via RFT in $T = 1$ s, these three targets are clearly located at (80 Km, 0 m/s), (80 Km, 300 m/s), and (80 Km, -300 m/s), respectively, which are all much higher than the noise level of Fig. 6(b).

2) $R_T = 150$ Km: According to radar equations, it is known that the input SNR of three targets are all -22.3 dB, and the targets’ raw echoes are below the noise level and unobservable. With the SNR gain $G_{\text{pc}} = 28.8$ dB of pulse compression, these three targets loom at the 150 Km range as Fig. 7(a). Furthermore, with 23 dB SNR gain via RFT in $T = 1$ s, there are three peaks that are clear at (80 Km, 0 m/s), (80 Km, 300 m/s), and (80 Km, -300 m/s), respectively, which are all much higher than the noise background and may easily be high-performance detected as Fig. 7(b).

3) $R_T = 300$ Km: According to the system parameters, it is known that the input targets’ SNR are -36.4 dB and obviously below the noise level. With the pulse compression SNR gain $G_{\text{pc}} = 28.8$ dB, the outputs are still below noise level and unobservable as Fig. 8(a). However, with 23 dB SNR gain via RFT in $T = 1$ s, there are three peaks that are clearly located at (80 Km, 0 m/s), (80 Km, 300 m/s), and (80 Km, -300 m/s), respectively, which are still higher than noise background and satisfy the needs of successive detection Fig. 8(b).

C. Improve the Weak Target Detection

To demonstrate the weak target detection performance of MTD, Hough-based method, and RFT, respectively, we give a scenario as follows. Assume there is a target located at range $R_T = 150$ Km with a maximum radial velocity $v_{\text{max}} = 2 \text{ Mach} \approx 680$ m/s. In order to detect the target with detection probability $P_d = 80\%$ with a given constant false alarm ratio $P_f = 10^{-6}$, the needed SNR should be 12.8 dB before binary decision. Then, the detectable minimum RCS for different methods can be obtained via (73)–(75)

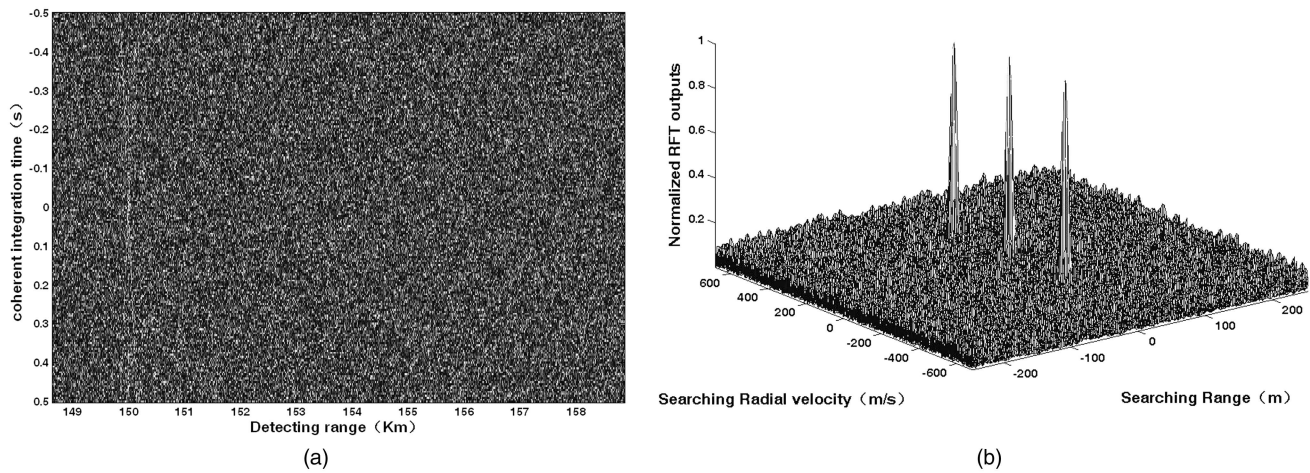


Fig. 7. Target detection at $R_T = 150$ Km. (a) After pulse compression. (b) RFT with $r_c = 150$ Km.

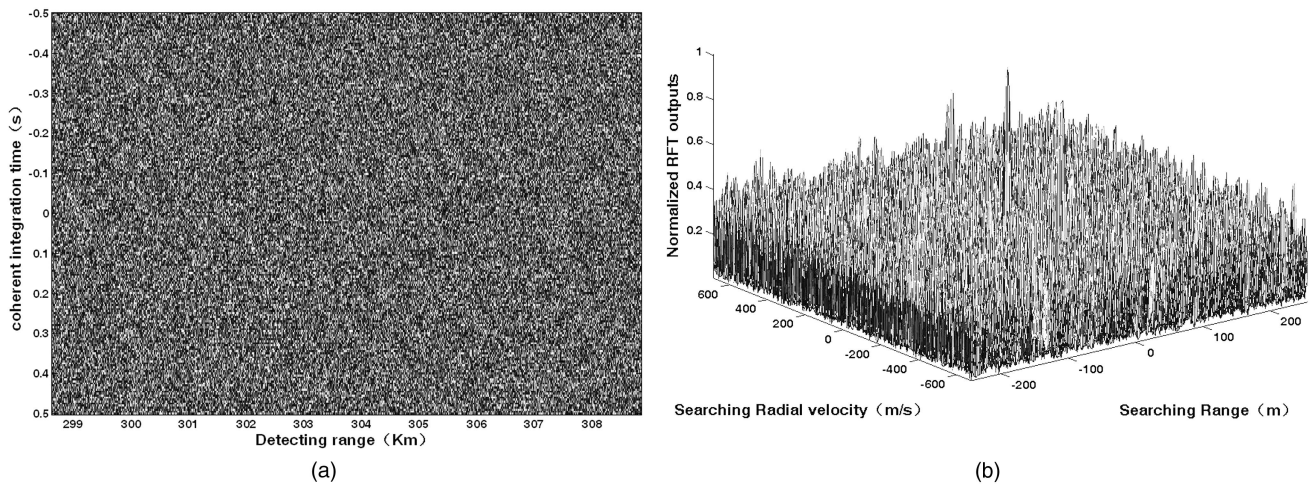


Fig. 8. Target detection at $R_T = 300$ Km. (a) After pulse compression. (b) RFT with $r_c = 300$ Km.

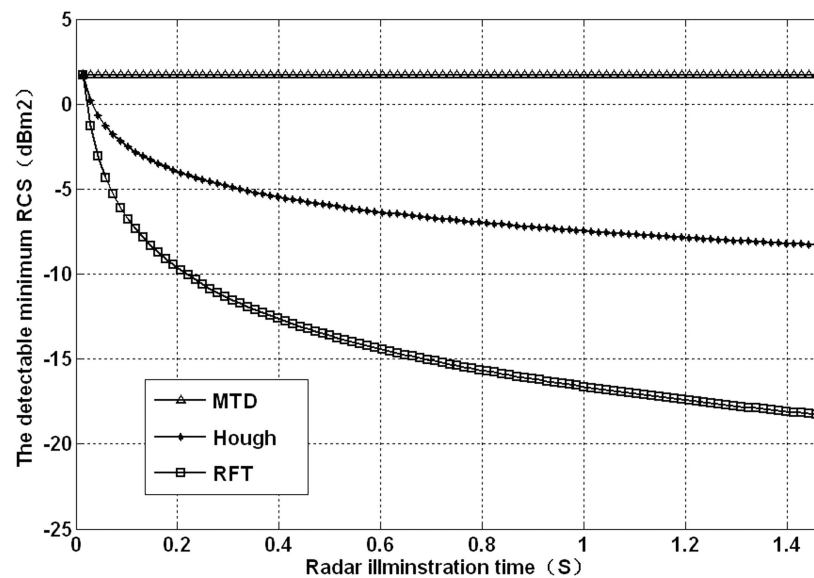


Fig. 9. Comparisons of minimum detectable RCS via three methods.

and the curves of $\sigma-T$ are given as Fig. 9. It is shown that the minimum detectable RCS via RFT at $T = 1$ s is approximately 0.02 m^2 , which is much smaller than

other methods. For example, the minimum detectable RCS of MTD may be larger than 1.47 m^2 , while that of the Hough-based method may be larger than

TABLE II
Maximum Detecting Range of Different Methods

Methods	MTD	Hough-Based Method	RFT
Detecting range (Km)	96.2	163.1	276.3

0.17 m² with hybrid incoherent-coherent integrating all the pulse samplings. In a word, the RFT may significantly improve the weak target detection in a given coherent integration time.

D. Increase the Detecting Range for the High-Speed Target

To demonstrate the maximum detectable ranges via MTD, Hough-based methods, and RFT, respectively, we design a simulation scenario as follows. Suppose the radar adopts the conventional mechanical scanning antenna to cover the 360° azimuth area with rotation velocity $\omega_r = 6$ rev/min with the given radar parameters as in Section VIIA, and the target is high speed and nonfluctuating with $v_{\max} = 2$ Mach ≈ 680 m/s. Then the specific steps are given as follows.

Step 1 Calculate the radar illumination time T_i and the integrated pulse number N_a as

$$T_i = \theta_\beta / \omega_r = 0.25 \text{ s}$$

$$N_a = T_i f_p = 50.$$

Step 2 Calculate the time of target pass through a range unit and the integrated pulses

$$T_M = \frac{c}{2Bv_{\max}} \approx 0.004 \text{ s}$$

$$N_{a,M} = \text{round}(T_M f_p) \approx 1.$$

Step 3 Calculate the maximum detectable range of MTD, Hough, and RFT as (76)–(78), respectively, and the Table II is obtained.

For the given high-speed target and the system parameters, it is shown that RFT may increase more than 180 Km range diagram more than the MTD and also increase about 113 Km range diagram more than the Hough-based method.

VIII. CONCLUSIONS

For the far-range, weak, high-speed target detection via coherent radar, a novel method, i.e., RFT, is proposed in this paper to realize the long-time coherent integration. RFT accomplishes the data extraction on the range-compressed signal with a 2D searching along range and radial velocity directions. Then, the Doppler filter bank is used for the successive long-time coherent integration. According to different searching parameters, four kinds of continuous and discrete RFTs are derived, respectively. Furthermore, a generalized form of

RFT, i.e., GRFT, is also defined for radar target detection with arbitrary parameterized motion. Also, this paper provides detailed comparisons between RFT and the existing MTD on five aspects, i.e., coherent integration time, filter bank structure, blind speed response, integration performance, and realization complexity. It is shown that MTD is actually a special case of RFT and the RFT is a kind of generalized Doppler filter bank. Finally, the detailed numerical experiments are provided to show the equivalence among four discrete RFTs. It is also validated that the RFT may obtain the ideal coherent integration gain in different SNR background. In the mean time, the proposed RFT method can significantly improve the weak target detection performance and increase the coverage diagram without change of system hardware.

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